Documentation

RealTransforms

Physics Placement for Unity

Documentation for physics-assisted placement, key assignment, toggling, performance tuning, collider baking, object baking, and troubleshooting.

Open	Window > RealTransforms	**Toggle**	CapsLock or Control tab
Tools	W Move, E Rotate, R Arrange	**Action**	Space
Package	Assets/RealTransforms - Physics Placement	**Unity**	6000.3.11f1



Version 1.0.0 documentation draft

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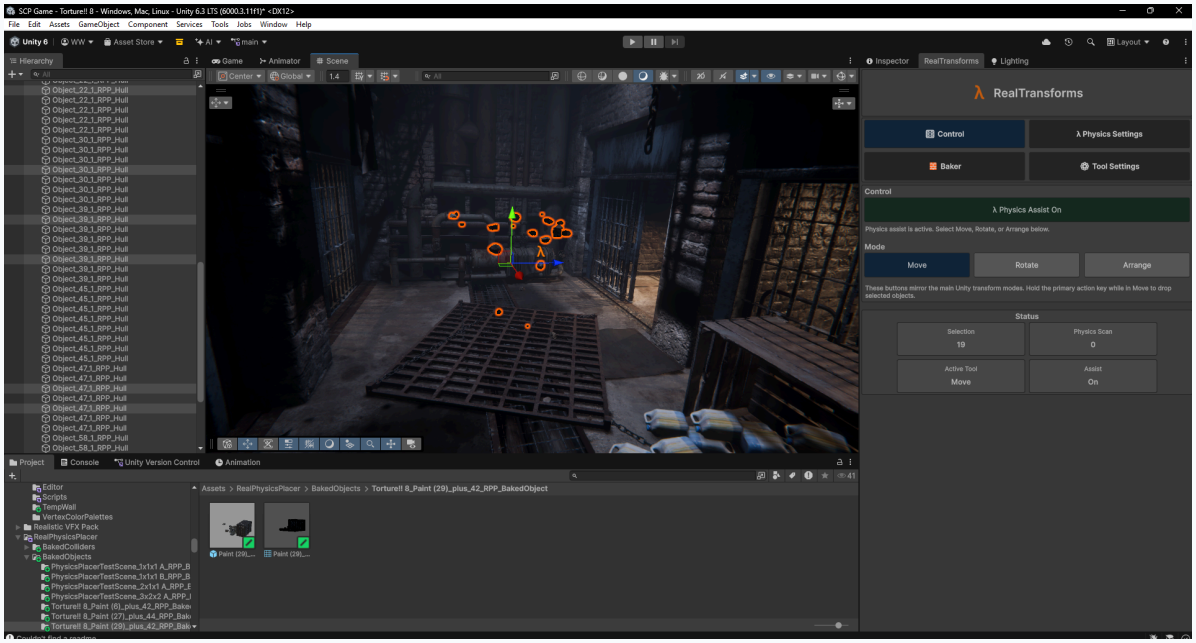
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Start Here

RealTransforms lets users use normal Unity transform muscle memory while temporary editor physics keeps objects from clipping, floating, or fighting contact.

1. Open Window > RealTransforms or Tools > RealTransforms > Open Window.
2. Select one or more scene objects.
3. Turn Physics Assist on with CapsLock or the Control tab button.
4. Use W for Move, E for Rotate, and R for Arrange.
5. Hold Space for the primary action. In Move this drops the selection. In Rotate or Arrange it pulls the selection upright while respecting contacts.

Best first run: Leave the default keys alone, choose the Balanced physics preset, and bake convex colliders for complex props before tight placement.



Controls And Toggling

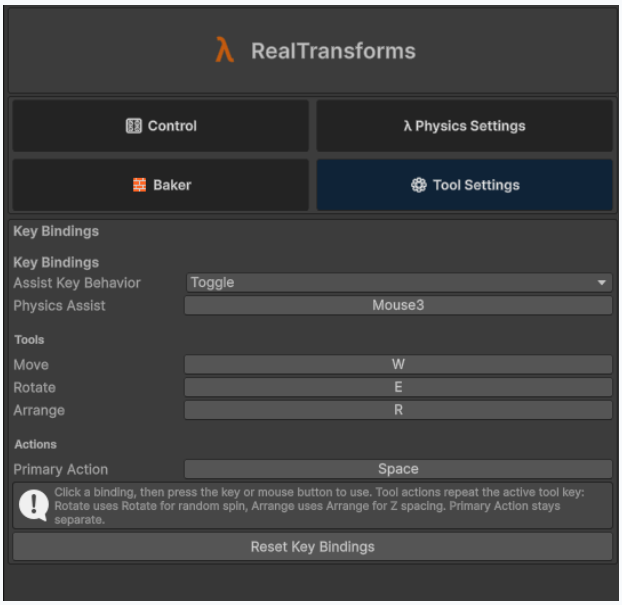
Key Assignments

Default Controls

Control	Default	What it does	Where to change it
Physics Assist	CapsLock	Toggles the whole editor physics overlay on or off.	Tool Settings > Key Bindings
Move	W	Physics-assisted Unity Move for placing objects against surfaces.	Tool Settings > Key Bindings
Rotate	E	Physics-assisted Unity Rotate for contact-aware turning.	Tool Settings > Key Bindings
Arrange	R	Physics-aware spacing for multiple selected objects.	Tool Settings > Key Bindings
Primary Action	Space	Drop in Move. Upright pull in Rotate or Arrange.	Tool Settings > Key Bindings

Reassigning Keys

- Open Tool Settings > Key Bindings.
- Click the binding button, then press the key or mouse button you want.
- Escape cancels key capture.
- Backspace or Delete clears the binding.
- Reset Key Bindings restores CapsLock, W, E, R, and Space.



Turning Physics Assist On And Off

Toggle Behavior

Action	Result	Note
Press CapsLock	Assist toggles on or off.	Default behavior is Toggle. It can be changed to Hold.
Click Physics Assist On/Off	Same master enable as the key.	Best for users who dislike keyboard toggles.
Enter Play Mode	Editor physics pauses.	The tool is intentionally editor-only and resumes after Play Mode.
Use W/E/R while assist is on	Switches the assisted mode.	Mirrors Unity Move, Rotate, and Scale/Arrange muscle memory.

Performance Overview

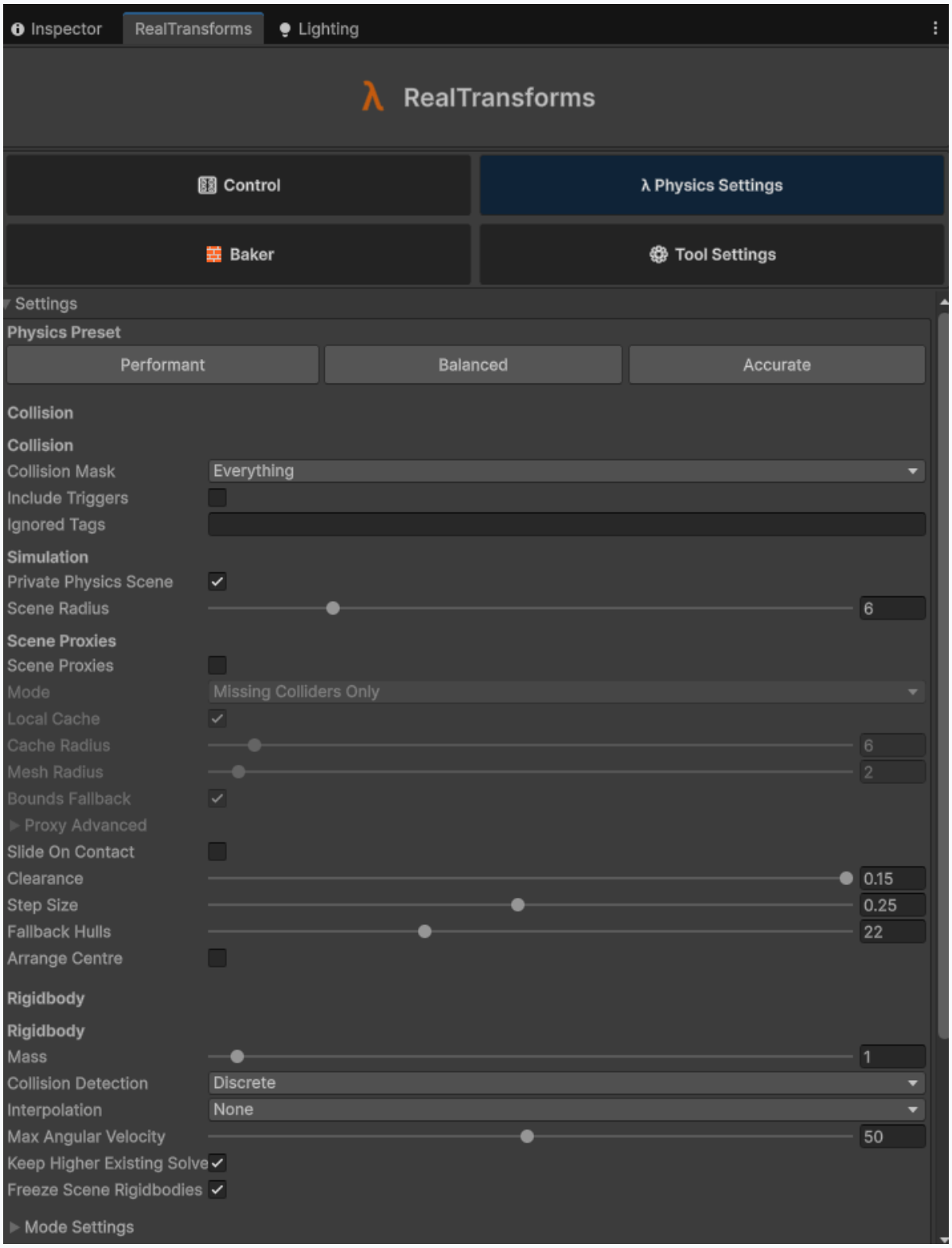
Most performance questions come down to four numbers: how many colliders are being scanned, how many baked hulls are active, how many physics steps run per editor update, and how much solver work each temporary Rigidbody does.

The Four Cost Levers

Cost lever	Watch this	Lower cost	Higher accuracy
Scene colliders	Physics Scan status and Scene Radius	Smaller radius, Performant, fewer scene proxies	Larger radius, Accurate, prefer mesh proxies
Baked hull count	Baked hulls and Max Hulls	Lower Max Hulls, simpler bake method	More hulls, tighter Real Hulls settings
Physics steps	Physics Step and Max Steps	Larger step, fewer max steps	Smaller step, more substeps
Solver work	Solver and Velocity Solver	Lower solver counts	Higher solver counts for stable contact/friction

Physics Preset Tradeoff Chart

Preset	Best for	Main settings	Tradeoff
Performant	Large scenes, fast blocking, low-end machines	Private physics on, scene proxies off, radius 6, fallback hulls 1, max steps 2	Fastest. Objects outside the private scene radius are not considered.
Balanced	Most projects	Private physics off, scene proxies off, radius 8, fallback hulls 1, max steps 2	Default-safe. Good responsiveness without scene-wide proxy scanning.
Accurate	Dense props, shelves, tight contact, tricky placement	Scene proxies on, mesh proxy radius 4, fallback hulls 4, max steps 4, Continuous Dynamic	Highest fidelity. More CPU, collider copying, and solver work.


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Everything
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Simulation
Private Physics Scene
Scene Radius
6
Scene Proxies
Scene Proxies
Mode
Missing Colliders Only
Local Cache
Cache Radius
6
Mesh Radius
2
Bounds Fallback
Proxy Advanced
Slide On Contact
Clearance
0.15
Step Size
0.25
Fallback Hulls
22
Arrange Centre
Rigidbody
Rigidbody
Mass
1
Collision Detection
Discrete
Interpolation
None
Max Angular Velocity
50
Keep Higher Existing Solve
Freeze Scene Rigidbodies
Mode Settings

Collider Baking

Collider baking is the cleanest answer for complex props. RealTransforms creates persistent convex compound colliders that behave well with Rigidbody physics. Use Smart Hulls for fast simple shapes and Real Hulls for more difficult concave meshes.

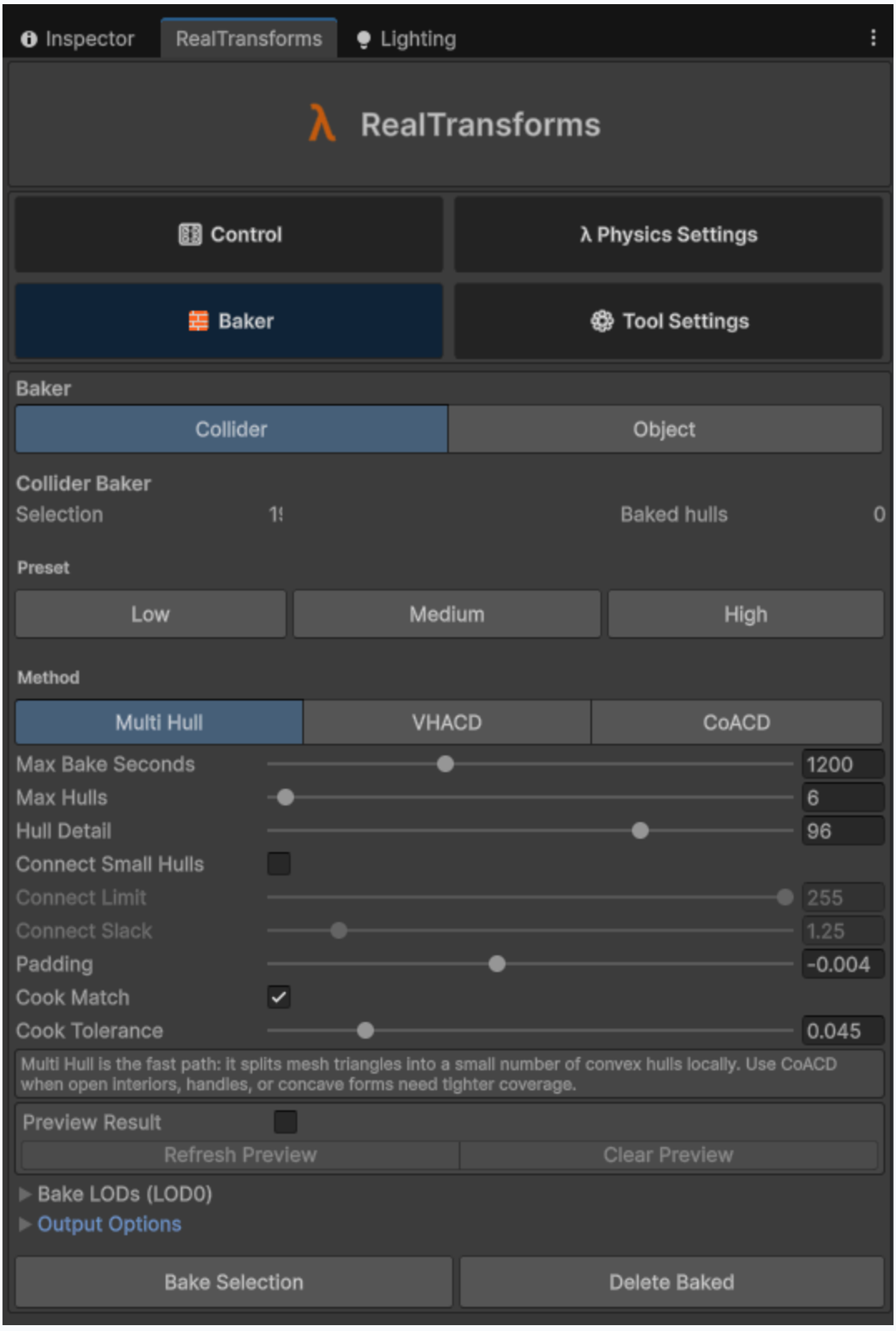
Collider Baker Tradeoff Chart

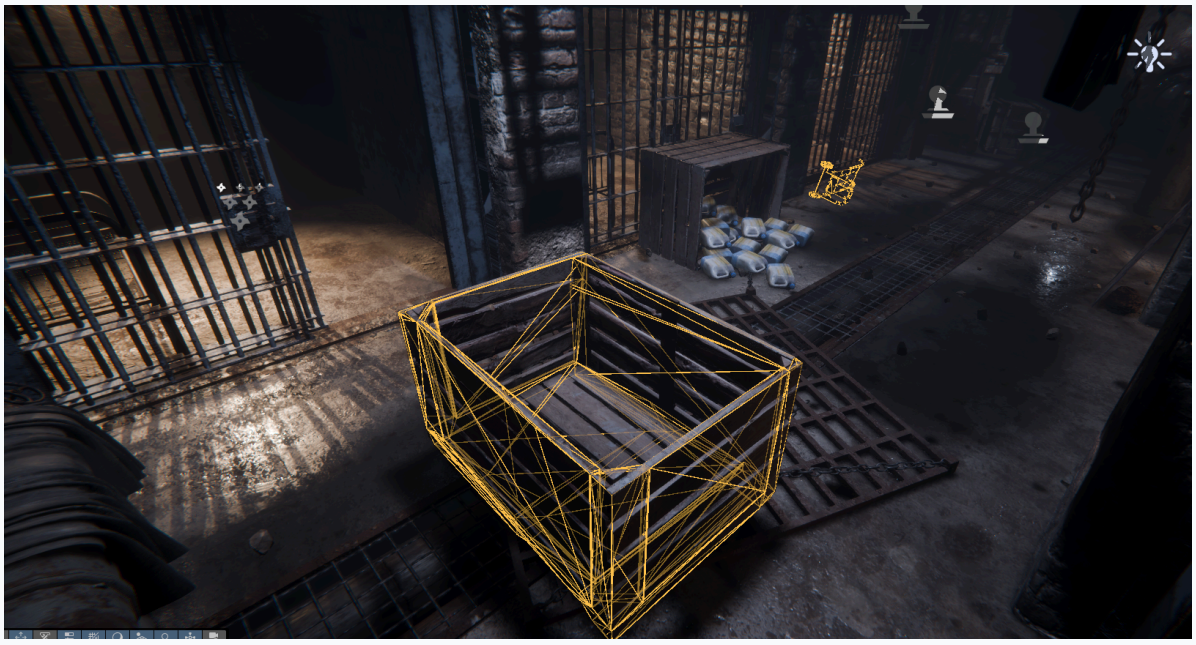
Method	Setup	Preview	Speed	Shape quality	Use it when
Smart Hulls	Built-in	Yes	Fast	Basic	Fast collider creation for simple or boxy props.
Real Hulls	Built-in beta solver	Manual button	Medium	Good	Complex props, hollow shapes, crates, shelves, and difficult concavity.
Burst Acceleration	Unity Burst package	Build-time only	Faster Real Hulls	Best	Install Unity Burst when you want faster Real Hulls bakes. The asset still works without it.

Collider Baking Workflow

6. Select the object or parent objects that need better physics placement.
7. Open Baker > Collider.
8. Set Max Hulls first. This is the biggest runtime cost control.
9. Set Hull Detail second. Unity convex MeshColliders are limited to 255 triangles.
10. Click Bake Selection.

Output: Baked colliders are saved under Assets/RealTransforms - Physics Placement/BakedColliders. Generated child roots are named __RealTransforms_BakedColliders, with Hull 001, Hull 002, and so on beneath them.

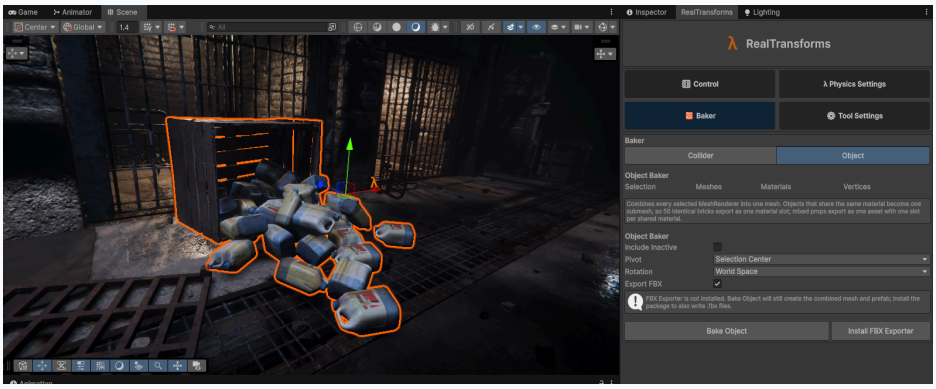

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Low
Medium
High
Method
Multi Hull
VHACD
CoACD
Max Bake Seconds
1200
Max Hulls
6
Hull Detail
96
Connect Small Hulls
Connect Limit
255
Connect Slack
1.25
Padding
-0.004
Cook Match
Cook Tolerance
0.045
Multi Hull is the fast path: it splits mesh triangles into a small number of convex hulls locally. Use CoACD when open interiors, handles, or concave forms need tighter coverage.
Preview Result
Refresh Preview
Clear Preview
Bake LODs (LOD0)
Output Options
Bake Selection
Delete Baked



Object Baking

Object Baker combines selected MeshRenderers into one optimized mesh and prefab. Shared materials become shared submeshes, which keeps repeated props tidy.

11. Select scene objects with MeshFilter and MeshRenderer components.
12. Open Baker > Object.
13. Choose Pivot and Rotation.
14. Keep Export FBX on if Unity's FBX Exporter package is installed.
15. Click Bake Object.



Settings Reference

The values below describe the included settings asset in this package. The Reset button restores script-defined product defaults.

Control And Input

Setting	Included value	Meaning
Assist Key Behavior	Toggle	Press once to turn on, press again to turn off.
Physics Assist Key	CapsLock	Master enable for editor physics assist.
Move Tool Key	W	Switches to physics-assisted Move.
Rotate Tool Key	E	Switches to physics-assisted Rotate.
Arrange Tool Key	R	Switches to physics-assisted Arrange and Arrange tool actions.
Primary Action	Space	Drop or upright action, depending on mode.

Collision And Scene Proxies

Setting	Included value	Meaning
Collision Mask	Everything	Layers that editor physics can collide with.
Include Triggers	Off	Trigger colliders are ignored unless enabled.
Ignored Tags	Empty	Comma-separated tags to ignore.
Private Physics Scene	Off	Optional isolated simulation. Faster in some scenes, but only sees colliders inside Scene Radius.
Scene Radius	8	Radius used by Private Physics Scene when it is enabled.
Scene Proxies	Off	Optional editor-only colliders for render-only scene geometry.
Proxy Mode	Missing Colliders Only	When Scene Proxies is enabled, only objects without usable colliders get proxies.
Local Cache	On	Keeps proxies near the selection instead of the whole scene.
Cache Radius	8	Distance where proxy colliders stay alive.
Mesh Radius	3	Distance where accurate static mesh proxies are allowed.
Bounds Fallback	On	Uses renderer bounds if no usable collider or mesh proxy exists.
Refresh Distance	1.5	Distance the selection moves before local proxies refresh.
Slide On Contact	Off	When off, blocked motion stays blocked instead of sliding along surfaces.
Clearance	0.15	Contact buffer to reduce visual clipping.
Step Size	0.25	Maximum movement slice used while checking collisions.
Fallback Hulls	4	Temporary selected-mesh fallback hull count when no collider exists.

Setting	Included value	Meaning
Arrange Centre	Off	Arrange uses pivot/Rigidbody behavior instead of forcing bounds center.

Rigidbody

Setting	Included value	Meaning
Mass	1	Mass for temporary editor Rigidbodies created during assist.
Collision Detection	Continuous Dynamic	Safer for fast contact-heavy movement, more expensive than Discrete.
Interpolation	Interpolate	Smoother visual motion, slightly more work.
Max Angular Velocity	50	Temporary Rigidbody spin cap.
Keep Higher Existing Solvers	On	Existing Rigidbodies keep higher solver counts than the active profile.
Freeze Scene Rigidbodies	On	Non-selected scene Rigidbodies are frozen while editor physics is active.

Move Mode

Setting	Included value	Meaning
Preserve Selection Offsets	On	Keeps selected objects spaced relative to each other.
Control Rotation	On	Contacts can rotate moved objects while placing.
Readjust Handle On Release	On	Re-anchors Unity's transform handle after mouse release.
Teleport When Too Far	Off	Keeps motion smooth instead of snapping back to the handle.
Move Max Speed	80	Maximum follow speed.
Simulation Speed	2	Time multiplier for move physics.
Physics Step	0.01	Fixed simulation step. Smaller is smoother and more expensive.
Max Steps	4	Maximum physics substeps per editor update.
Solver	12	Position solver iterations.
Velocity Solver	8	Velocity/friction solver iterations.
Move Spring	240	Pull strength toward the handle.
Move Damping	24	Linear damping while following.
Move Spin Damping	32	How quickly unwanted rotation calms down.
Move Smoothing	0.35	Mouse-handoff smoothing time.
Move Leash	1	Maximum distance from handle to physics target.
Contact Assist	0.1	Distance before move physics engages near collision.

Rotate Mode

Setting	Included value	Meaning
Simulation Speed	2	Time multiplier for rotate physics.
Physics Step	0.008	Fixed rotate simulation step.
Max Steps	4	Maximum rotate substeps per editor update.
Solver	16	Position solver iterations.
Velocity Solver	8	Velocity solver iterations.
Turn Pull	90	Rotation spring strength.
Turn Damping	16	Rotation damping.
Turn Max Speed	14	Maximum angular speed.
Slide Damping	2.5	Sideways drift damping while rotating against contacts.
Turn Step	5	Maximum degrees applied per rotation solve.
Contact Tolerance	0.04	Buffer that helps prevent sinking into nearby surfaces.

Drop Mode

Setting	Included value	Meaning
Gravity	(0, -9.81, 0)	Drop gravity vector.
Simulation Speed	1	Drop time multiplier.
Physics Step	0.01	Fixed drop simulation step.
Max Steps	4	Maximum drop substeps per editor update.
Solver	8	Position solver iterations for dropped Rigidbodies.
Velocity Solver	6	Velocity solver iterations for dropped Rigidbodies.
Drop Passes	2	Extra placement passes for stable contact.
Max Speed	13.5	Maximum fall speed.
Settle	0	Extra quiet simulation steps after release.

Arrange Mode

Setting	Included value	Meaning
Handle Strength	2	Strength used when arrange handles push or pull objects.
Spacing	0.5	Extra space after object bounds touch.
Space Action	Grid XZ	Primary action behavior while Arrange is active.
Arrange Key Action	Clump	Behavior when Arrange is pressed again while Arrange is active.
Physics Aware	On	Arrangement uses collision checks.

Setting	Included value	Meaning
Selected Collide	On	Selected objects collide with each other while arranging.

Collider Baker

Setting	Included value	Meaning
Method	Real Hulls	Current selected bake method.
Time Limit	60	Time limit for one bake run.
Max Hulls	6	Maximum generated hulls per mesh. Lower is faster.
Hull Detail	128	Target vertices per hull, clamped to Unity-safe ranges.
Bake LODs	Lowest	LOD renderer levels included when an object has an LOD Group.
Enable Preview	On	Allows Smart Hulls and Real Hulls previews. Click Refresh Preview to rebuild; sliders do not auto-update the preview.
Padding	0	Signed hull padding. Positive expands, negative tightens.
Cook Match	On	Splits hulls that Unity cooks too far from the generated shape.
Cook Tolerance	0.035	Allowed local-space cook difference before splitting.
Connect Small Hulls	Off	Consolidates small neighboring hulls at bake time when enabled.

Real Hulls Settings

Setting	Included value	Meaning
Preset	Fast	Quality multiplier. Fast is the current default for responsive collider bakes.
Voxel Resolution	96	Internal occupancy grid resolution. Higher values preserve more concavity but cost more.
Surface Samples	11000	Maximum source samples used for scoring and hull fitting.
Split Candidates	64	Candidate split planes tested per recursive split.
Shape Accuracy	0.03	Lower values preserve more concavity; higher values are faster and looser.
Merge Pass	On	Merges compatible final hulls when the combined shape still matches the source.
Cache Results	On	Reuses identical Real Hulls results for repeated previews and bakes.
Cook Match	On	Checks Unity's convex cook and splits hulls that drift too far.
Beta Status	Beta	Please report meshes that Real Hulls does not represent accurately, including settings used.

Optional Acceleration

Setting	Included value	Meaning
Burst Package	Optional	RealTransforms works without it. Installing Unity's Burst package is recommended for faster Real Hulls bakes.
Job System	Optional	Used automatically for supported scoring paths when Burst support is available.
Managed Fallback	Included	Used automatically when Burst is not installed.
Result Quality	Same target	Acceleration changes bake time, not the intended collider shape.
Install Location	Package Manager	Add Unity Burst from Package Manager when faster bakes are needed.
Runtime Need	None	Baked colliders are normal Unity colliders.
Best Use	Large meshes	Most useful on expensive Real Hulls bakes.
Support Note	Report slow bakes	Include mesh, settings, and Unity version when asking for help.
Cache Results	On	Reuses identical Real Hulls input results.

Output Options

Setting	Included value	Meaning
Replace Baked	On	Deletes old baked collider children before rebaking.
Disable Existing	Off	Leaves existing colliders enabled unless changed.
Include Inactive	Off	Inactive child meshes are skipped unless enabled.
Skip Primitives	On	Unity primitive meshes keep their simple built-in collider path.
Add Mesh Filters	Off	Keeps generated hull children collider-only unless enabled for inspection.
Draw Gizmos	Off	Shows baked collider outlines when enabled.
Always Draw	Off	Keeps baked collider gizmos visible for unselected objects when Draw Gizmos is enabled.
Solid Preview	Off	Filled translucent hull preview is off by default.

Object Baker

Setting	Included value	Meaning
Include Inactive	Off	Inactive MeshRenderers are skipped.
Export FBX	On	Exports FBX when Unity's FBX Exporter package is installed.
Pivot	First Selected	Baked asset origin uses the first selected object's position.
Rotation	Average Direction	Baked asset direction blends selected root directions.

Troubleshooting

Common Pain Points

Symptom	Fix
CapsLock does nothing	Click the Scene view, make sure Unity is not in Play Mode, and check Tool Settings > Key Bindings. Non-CapsLock keys are ignored while typing in text fields.
Assist is on, but regular Unity tools still show	Select Move, Rotate, or Arrange in the RealTransforms Control tab, or press W, E, or R while assist is on.
Placement misses renderer-only scenery	Turn on Scene Proxies and Bounds Fallback, or switch to Balanced/Accurate.
Editor gets slow in a big scene	Use Performant, reduce Scene Radius, keep Local Cache on, reduce Max Hulls, and bake fewer collider hulls.
Object clips through tight geometry	Use baked convex colliders, lower Physics Step, raise Max Steps, and try Accurate.
Real Hulls preview does not update	This is intentional. Change settings, then click Refresh Preview to rebuild manually.
Real Hulls result looks inaccurate	Try a higher Max Hulls, lower Shape Accuracy, higher Voxel Resolution, or report the mesh and settings.
Real Hulls bake is slow	Use the Fast preset, keep Lowest LOD, reduce Max Hulls, enable Cache Results, or install Unity's Burst package.
Convex MeshCollider triangle warning	Lower Hull Detail, increase Max Hulls, and keep Cook Match enabled.
Existing colliders fight baked colliders	Enable Disable Existing before baking, or manually disable old colliders after verifying the generated hulls.

Package Structure

Folders

Folder	Purpose
Editor	RealTransforms window, scene tool, physics assist, baker UI, Smart Hulls, Real Hulls, object baking, and editor-only helpers.
Runtime	Runtime-safe baked collider marker and gizmo component.
BakedColliders	Generated convex hull mesh assets.
BakedObjects	Generated combined mesh assets and prefabs.
Scenes	Example scene.
Materials	Demo materials.
Documentation	Documentation, screenshots, and setup notes.
Real Hulls Solver	Built-in clean-room C# collider generation. No external solver setup is required.

Asset Store Copy

RealTransforms gives Unity creators physics-assisted Move, Rotate, Arrange, and Drop tools directly in the Scene view. Place props against real colliders, drop objects into natural positions, arrange groups with collision awareness, and bake convex compound colliders for difficult meshes with Smart Hulls and Real Hulls.

Built for production scenes, RealTransforms includes performance presets, optional scene proxies, a local proxy cache, collider bake presets, Smart Hulls, Real Hulls beta collider generation, optional Burst acceleration, object baking, and clean runtime-safe baked collider assets.